

Ronit Baniya Gupta

Computer Engineer & Builder of Things

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PROFILE

Computer engineer from Pokhara, Nepal. I work with microcontrollers the way other people work with keyboards - and I've been at it since the first time I held an Arduino and was immediately done for. My work lives where code touches real wires: embedded systems, IoT ecosystems, computer vision and the occasional website when the pitch lands. I'd rather ship a rough working thing today than a perfect thing next year. I've taken hardware from breadboard to market - a home-automation switch with a custom PCB, a call-based motor controller that works where WiFi doesn't reach, and an emergency-response ecosystem that's swept the hackathon circuit. If it doesn't exist, I build it.

EDUCATION

- : Bachelor of Computer Engineering - Pokhara University
- : Diploma in Computer Engineering - Batch 2020
- : 3rd DAN Black Belt - Taekwondo
- : High-Performance Coach Trained
- : 3+ Innovation Awards

SKILLS

Languages: Python, C, C++, Java, JavaScript, HTML, CSS, VHDL

Hardware: ESP32, ESP8266, Arduino Uno, Arduino Nano, nRF52840, STM32

Sensors: MPU6050, MAX30102, Pulse Sensor, DS18B20, MLX90614, Ultrasonic, PIR, RFID, IR, GPS, SIM800L

Displays: OLED, LCD I2C, 1.28" Round TFT

Comms: HC-05, HM-10, NRF Modules, BLE, Wi-Fi, MQTT

Power: 18650, TP4056, Buck Converters, LM2596, 7805, Bridge Rectifiers

Software: Firebase, Supabase, React, Node.js, Flutter, OpenCV, MediaPipe, PlatformIO, ESP-IDF, MIT App Inventor

PROJECTS

- : AEGIS (2026) - Disaster response ecosystem (concept)
- : Lyrics Display Systems (2026) - OLED & LCD lyric players
- : Fishtail (2026) - Dev-core + hardware workspace
- : Ghostware360: Debug or Die (2026) - Horror puzzle game, in progress
- : Bulk QR Generator (2026) - Hundreds of QR codes, one script
- : USB Rubber Ducky builds (2026) - Keystroke injection experiments
- : WiFi Tracking & CSI (2026) - RuView + channel-state experiments
- : ESP Utility Toolkit (2026) - MAC finder, pin checker, tag victim
- : Earthquake Simulator (2026) - Python/Flask app that shakes
- : Mass Report (TikTok Bot) (2026) - Automation for a reason
- : AI Agents & Hand Detection (2026) - REAL_AGENT, AI AGENT & vision demos
- : A.I.R.I.S. (2026) - Personal voice assistant
- : JAVA Practical (2026) - Semester Java practice
- : ESP-IDF Hello World (2026) - Day 01 / Test / hello_world experiments
- : PahiloMart WebPage (2026) - Vite + TypeScript storefront
- : Self-Hosted n8n (2026) - Workflow automation, my own server
- : Laser Tag (2026) - IR tagger system, showcased at the expo
- : Roundabout with Pedestrian Bridge (Prithvichowk) (2026) - Civil model - first runner-up
- : SwiftIgnite v2 (2026) - Live GPS tracking + SIM800L telemetry
- : Suraksha360 (2025) - Smart security & emergency response ecosystem
- : SwiftIgnite (2025) - Keyless motorcycle ignition system
- : RUDRA-I Rocket (2025) - Experimental rocket with telemetry & recovery
- : GRAVAS (2025) - Gesture-controlled robotic arm

- : Swasthya Sathi (2025) - Healthcare platform with AI symptom checker
- : Call-Based Motor Control (2025) - Turn motors on/off with a phone call
- : Restaurant Websites (ROS) (2025) - Butter Mellow, Ember & Ivy, Nako Cafe & more
- : D:app - Next.js App (2025) - App router experiments
- : Assembly Level Coding (2025) - Talking to the metal
- : Computer Graphics (2025) - Semester graphics course work
- : SurakshaGhar (2024) - Home automation I actually shipped
- : Smart Motor Control System (2024) - Motor control circuit, designed in Proteus
- : Smart Parking System (2024) - Parking that tells you where to go
- : GPS Tracker & Buck Converter (2024) - Track a thing, power it properly
- : SlideSmart (2024) - Mobile-controlled curtain automation
- : Custom Buck Converter (2024) - Down-voltage, my way
- : Custom Relay Module (2024) - Relay board, designed from scratch
- : Home Motor Timer (2024) - Motor on for 6 hrs, off for 1 min, repeat
- : M.A.M.A (2024) - Multi-Agency Management Assistant
- : HerLegal (2024) - Legal resource site
- : Diploma CTEVT Checker (2024) - Result checker desktop app
- : Cloned & Scraped Websites (2024) - royevent.com, GandakiTent, SlayerCloudSite & more
- : Personal Website (Next.js) (2024) - The first version of this site
- : Project05 PA (2024) - Home-lab mini project
- : College Semesters (PEC) (2024) - BE.PEC & PEC semester folders
- : Home Automation Lab (2023) - ESP32 experiments: LDR, ultrasonic, relays
- : SmartDustBin (2023) - A dustbin with a brain
- : RC Car Builds (2023) - Rc_Car1, Rc_Car2 & the NRF stealth version
- : RoomLightApp (2023) - Room lighting from your pocket
- : Group Minor Project (PHP) (2023) - Login/signup system
- : Air Mouse (2022) - Mouse controlled by pointing

ACHIEVEMENTS

- : 1st Position - UDAAN 2025 - Project Expo (IOE Paschimanchal Campus, WRC) (2025)
- : Certificate of Appreciation - UDAAN 2025 Project Expo (2025)
- : People's Choice Award - OCTATE 2025 - National Level Technical Festival (2025)
- : First Runner-Up - Innosphere 1.0 - Hardware Project Exhibition (2025)
- : Appreciation - CAPEC (2025) - Leadership & Dedication
- : Appointed - CAPEC (2026) - Joint Secretary, 2026
- : Consolation - 9th PEC Tech Expo 2026 - Computer & IT Project Competition (2026)
- : Participant - 9th PEC Tech Expo 2026 (2026)
- : First Runner-Up - 9th PEC Tech Expo 2026 - Civil Project Competition (2026)
- : Community Partner - Hackathon Nova (2026)
- : Volunteer - Hackathon Nova Volunteer (2026)
- : Presented - Fishtail University Fair (2026)
- : 1st Place - GCES IT Expo 2026 (2026)
- : Participated - Codefest 2026 - Gandaki Province (2026) - Team Suraksha360

PRINCIPLES & EXTRAS

- : Originality over everything - No copy-paste ideas. If someone already made it, I make it better or I don't make it.
- : Build it, then improve it - I'd rather ship a rough working thing today than a perfect thing next year.
- : Brutally honest feedback - Tell me I'm wrong and why. Comforting lies never built anything.
- : 3rd DAN Black Belt in Taekwondo - started BS 2070, stopped, came back, finished high-performance coach training.
- : Weak at theory, strong at shipping. I learn better with a screwdriver in my hand.